# SWA-DPBR: Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability for Cross-Cultural Moral Alignment of Large Language Models

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## Abstract

Large language models make moral judgments as if the whole world thinks like the West-yet 40 million human responses across 233 countries say otherwise. We show that a frozen LLM can be taught to *respect* that diversity *without any training*. We introduce **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability), a training-free, inference-time method that steers binary moral choices toward country-specific human preferences using only logit corrections-no weight updates, no per-country finetuning, no white-box access. The key insight is that *disagreement is signal*: we instantiate each country as a panel of World Values Survey–grounded persona agents whose within-country disagreement pinpoints exactly where and how much to correct the model's default. That structured disagreement is converted into a bounded logit shift via Prospect-Theory importance sampling, stabilised through a dual-pass bootstrap reliability gate that automatically shrinks corrections when sampling is noisy. Evaluated on MultiTP across **twenty countries, four continents**, and six open model families (3.8B–70B parameters), SWA-DPBR reduces misalignment by **19–24%** on the strongest checkpoints, with gains spanning the Americas, Europe, and Asia-not confined to any single region. When the method fails, the culprit is architectural—poorly conditioned decision logits rather than cultural blind spots—suggesting that inference-time cultural steering is broadly viable wherever the base model maintains a well-behaved decision surface. Code: [repository].

## 1 Introduction

A self-driving car must choose: spare a group of elderly men or a group of young women. Ask in Tokyo and Bogotá and you will get systematically different answers. The Moral Machine experiment [**?**] demonstrated this at scale: 40 million judgments from 233 countries revealed that moral preferences are not noise—they are *structured* by culture. East Asian respondents weight youth more heavily; Southern-hemisphere populations show distinct utilitarian calculi; Islamic societies prioritise different social roles. Yet the large language models increasingly enlisted to mediate such decisions encode a single, culturally monolithic moral stance inherited from predominantly Western training data [**?**]. The multilingual MultiTP benchmark [**?**] now quantifies the scale of this gap: country-specific Average Marginal Component Effects (AMCE) across six moral dimensions and 107 languages show that *none* of 19 contemporary LLMs achieves broadly satisfactory alignment, with most exhibiting WEIRD-biased reasoning that departs sharply from non-WEIRD human preferences.

Closing this gap without retraining is hard because a practical cross-cultural decoder must satisfy three constraints at once: no weight updates, no per-culture supervised labels at scale, and no white-box access to internal activations. Every existing approach trades at least one away—fine-tuning [**??**]

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changes weights and needs per-culture data; prompting-based steering lacks empirical grounding [**?**]; activation steering [**??**] requires internal-layer access; reward-guided decoding [**??**] assumes per-country reward models that do not exist; and culture-aware adapters [**??**] need per-culture training, scaling poorly past a handful of countries.

We propose **SWA-DPBR** (Socially-Weighted Alignment with Dual-Pass Bootstrap Reliability), a method that satisfies all three constraints by exploiting a simple but powerful insight: *within-country disagreement is the steering signal itself*. Moral preferences within a society are not monolithic—age cohorts and social strata prioritise fundamentally different trade-offs [**??**], and the World Values Survey [**?**] provides empirical structure for that variation. SWA-DPBR instantiates each country as a small panel of culturally grounded persona agents drawn from WVS microdata; when those agents disagree on a dilemma, the spread of their logit gaps pinpoints where a correction is needed and how large it should be, and when they agree, the vanilla model is typically already close and no correction is applied. Operationally, the pipeline adds a single batched forward pass (base model plus personas in the native language), applies positional debiasing, then runs two independent Prospect-Theory importance-sampling passes [**?**] whose outputs are merged with a bootstrap reliability weight that automatically shrinks corrections when the two passes disagree. An effective-sample-size guard and a light hierarchical country prior complete the stack (§3; hyperparameters in Table 7).

**Contributions.**

- **A training-free cultural alignment method.** SWA-DPBR aligns a *frozen* LLM to country-level human preference vectors at inference time: WVS-grounded personas supply structured disagreement; Prospect-Theory IS converts it to a bounded logit correction; dual-pass bootstrap reliability gates noisy updates; a hierarchical prior smooths across scenarios (§3).
- **Broad empirical validation.** On MultiTP across twenty countries, four continents, and six model families (3.8B–70B), SWA-DPBR cuts misalignment by 19–24% on the strongest checkpoints, with gains that span geographies rather than concentrating in one region (§5).
- **Diagnostic ablations.** We show that persona diversity, positional debiasing, and the PT kernel are each load-bearing; that temperature and preprocessing sweeps leave conclusions stable; and that failure traces to collapsed logit entropy, not cultural blind spots (§6; Appendices E–F).

## 2 Related Work

**Benchmarks and cultural evaluation.** The Moral Machine experiment [**?**] established that moral preferences are culturally structured; MultiTP [**?**] extends this to 107 languages with country-specific AMCEs, providing the evaluation bed we adopt. Existing moral benchmarks (ETHICS [**?**], MoralExceptQA [**?**], MoCa [**?**]) remain English-centric, and broader diversity efforts [**?????**] cover fewer languages or different modalities. We validate on MultiTP's binary trolley vignettes where AMCE vectors provide a clean six-dimensional surface; a pilot extension to the open-ended BLEnD benchmark [**?**] (Appendix N) reveals that the binary logit-gap formulation does not transfer directly to free-form generation, and no existing training-free method has demonstrated country-level calibration on such tasks. On the diagnostic side, LLM outputs correlate most strongly with WEIRD respondents [**?**], RLHF introduces unintended cultural shifts [**?**], and recent work shows that LLMs systematically stereotype the moral values of non-Western populations [**?**]—underscoring the practical need for culturally grounded correction mechanisms.

**Inference-time alignment.** The pluralistic alignment vision [**?**] argues that no single preference ordering serves all communities; inference-time methods are a natural vehicle because they avoid baking a fixed value stance into weights. However, existing approaches each sacrifice one of the three constraints we target: activation steering [**??**] requires white-box access; reward-guided decoding [**??**] needs per-country reward models; and culture-aware adapters [**??**] need per-culture training. Recent logit-level steering work [**?**] and culturally localised generation via sparse autoencoders [**?**] demonstrate that the logit surface is a richer control interface than previously appreciated, but neither targets country-conditioned moral calibration. Inference-time stochasticity can improve moral alignment by reshaping confidence [**?**]; SWA-DPBR complements this by targeting *directional*, survey-grounded corrections rather than generic uncertainty inflation. To the best of our knowledge, SWA-DPBR is the first training-free method targeting country-level moral calibration specifically.

**Theoretical grounding.** SWA-DPBR draws on the control-as-inference view from path-integral steering [**??**], where softmax-weighted perturbations approximate the optimal control distribution. We shape the utility with Prospect Theory [**?**] to encode loss aversion—penalising corrections that worsen any persona's alignment more heavily than rewarding equal-sized gains. Recent evidence that prospect-theory biases emerge naturally in LLM decision-making [**?**] lends additional motivation: the asymmetric gain/loss structure we impose mirrors one the models themselves already exhibit, rather than introducing an alien objective. The dual-pass reliability gate splits the Monte Carlo budget into two independent replicates to detect sampling noise without doubling forward passes; we treat it as a variance heuristic rather than a formal unbiasedness certificate (Appendix C). Structured multi-agent systems such as OG-MAR [**?**] pursue richer ontology-level reasoning; we trade that depth for a single additional batched forward pass and explicit country-level calibration.

# 3 Method: SWA-DPBR

## 3.1 Problem Formulation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` ($a$) and `B` ($b$). For target country $c$, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human AMCE vector from 40M+ judgments [**?**]. A frozen LLM $f_\theta$ produces logits $\mathbf{z}(\mathbf{x}) = [z_a,\ z_b]$ at the decision token position; the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ where $\Delta = z_b - z_a$. Each dimension's AMCE is estimated as:

$$\hat{m}_d^{(c)} = \frac{1}{|\mathcal{S}_d^{(c)}|} \sum_{\mathbf{x} \in \mathcal{S}_d^{(c)}} p_{\text{spare}}(\mathbf{x}), \tag{1}$$

following the MultiTP evaluation convention [**?**]. For the five binary dimensions (Species, Gender, Age, Fitness, Social Value), this reduces to the empirical mean of $p_{\text{spare}}$. Utilitarianism requires a linear regression on the count difference between groups (Appendix A).

**Goal:** Without modifying $\theta$, find a logit correction $\delta^*(\mathbf{x}, c)$ such that the corrected AMCE vector $\hat{\mathbf{m}}^{(c)}(\theta, \delta^*)$ is closer to $\mathbf{h}^{(c)}$. We primarily measure alignment with the *misalignment score* (MIS): $\ell_2$ distance between $\hat{\mathbf{m}}^{(c)}$ and $\mathbf{h}^{(c)}$ (lower is better). Jensen–Shannon divergence and Pearson $r$ are reported as secondary diagnostics where noted.

## 3.2 Cultural Persona Construction

For each country $c$, SWA-DPBR constructs $N = 4$ persona agents from WVS Wave 7 individual-level microdata [**?**], following the 10-variable cultural-value scheme of Greco et al. (2025). We extract: *religiosity* (Q6P), *child-rearing values* (Q17P), *moral acceptability* (Q177–Q182), *social trust* (Q57P), *political participation* (Q199P, Q200P), *national pride* (Q254P), *happiness* (Q46P), *gender equality* (Q29P–Q33P), *materialism orientation* (Q152–Q153), and *tolerance for diversity* (Q19P–Q23P). Items from the "inverted" WVS-7 CSV (suffix P) are directionally coded so that higher values consistently indicate the positive pole of each dimension. For each country, respondents are segmented into three age cohorts using birth year and survey year: *young* (<36), *middle-aged* (36–55), and *older* (>55). Dimension means are normalised to $[0, 1]$ and mapped to four-level natural-language descriptors via quartile cuts ($\geq 0.75$, $\geq 0.50$, $\geq 0.25$, $< 0.25$); see Appendix B. A fourth, country-invariant *utilitarian agent* maximises lives saved. Each persona is a native-language system prompt assembled from a header, one sentence per WVS dimension with data, and a moral-reasoning closing line.

WVS grounding ensures empirically measured within-country variation rather than stereotypes [**?**]. Concrete persona examples and the WVS-to-trolley dimension linkage are detailed in Appendix B.

## 3.3 Batched Evaluation and Logit Extraction

All $N + 1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\ \mathbf{z}_1, \ldots, \mathbf{z}_N = f_\theta(\mathbf{x};\ \varnothing,\ \mathbf{s}_1, \ldots, \mathbf{s}_N)\,. \tag{2}$$

Per-category logit temperatures $T_{\text{cat}}$ scale logits before computing decision gaps. We use $T_{\text{cat}} = 4.0$ (Species), $3.5$ (Gender), and $1.5$ for Age, Fitness, Social Value, and Utilitarianism (Table 1); the same

values appear in Appendix D.

$$\delta_i = \frac{z_{i,b} - z_{i,a}}{T_{\text{cat}}}, \quad \delta_{\text{base}} = \frac{z_{\text{base},b} - z_{\text{base},a}}{T_{\text{cat}}}, \quad \bar{\delta} = \frac{1}{N}\sum_{i=1}^{N} \delta_i \quad \text{(consensus)}, \tag{3}$$

$$r_i = \delta_i - \delta_{\text{base}} \quad \text{(per-agent reward)}. \tag{4}$$

Here $\delta_{\text{base}}$ is the decision gap of the base prompt (no persona). The reward $r_i$ measures how strongly persona $i$ shifts the model *away from the culturally-neutral baseline*: a large $|r_i|$ indicates that the persona has identified a direction of moral preference the vanilla model does not exhibit. (Note: the scalar $r$ introduced later in Eq. 6 denotes the bootstrap reliability weight, not a reward.) Inter-persona sample variance is logged as a diagnostic but **does not gate** importance sampling in the released implementation (no $\tau$ threshold on inter-persona variance).

### 3.4 Positional Debiasing

To cancel recency bias [**?**], each scenario is evaluated twice: once in the original ordering, once with *both* choice labels (A $\leftrightarrow$ B) and group labels (Group A $\leftrightarrow$ Group B) swapped in the native language. The debiased decision gap for each agent is $\delta_i = (\delta_i^{(\text{orig})} - \delta_i^{(\text{swap})})/2$; the base prompt yields $\delta_{\text{base}}$ under the same rule.

### 3.5 From Persona Disagreement to a Bounded Correction

The persona consensus $\bar{\delta}$ already captures the cultural direction, but applying it directly would ignore two problems: the consensus may overcorrect toward one loud persona, and finite-sample noise may push the correction in an unreliable direction. We address both through importance sampling with a cooperative, loss-averse utility.

The idea is to *search* for a correction that simultaneously improves alignment with every persona, not just the loudest one. We sample candidate perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ around the consensus and score each candidate by how much it moves the model closer to each persona relative to the uncorrected baseline. These per-persona gains are aggregated through a Prospect-Theory value function that treats improvements and regressions asymmetrically—a correction that *helps* three personas but *hurts* one is penalised more heavily than a correction that helps all four modestly. This "do no harm" prior is the core mechanism that prevents the IS stage from overcorrecting:

$$U_{\text{total}}(\epsilon_k) = (1 - \lambda_{\text{coop}})\, \frac{1}{N}\sum_{i=1}^{N} v\Big(\frac{g_{i,k}}{\sigma}\Big) \;+\; \lambda_{\text{coop}}\, v\Big(\frac{g_{\text{cons},k}}{\sigma}\Big), \tag{5}$$

where $v(\cdot)$ is the Kahneman–Tversky value function with gain curvature $\alpha$=0.88 and loss aversion $\kappa$=2.25 [**?**], $g_{i,k}$ measures how much candidate $k$ improves over the base for persona $i$ (Appendix C), and $\lambda_{\text{coop}}$=0.7 weights the group consensus. Candidates are then softmax-weighted by utility and combined into a single correction $\delta_m^\star$; an effective-sample-size guard zeros the correction when the weights collapse, falling back on the raw consensus (Appendix C).

The second problem—sampling noise—is handled by the *dual-pass reliability gate*. Rather than trusting a single IS estimate, SWA-DPBR splits the budget into two independent passes and asks: do they agree? If yes, the correction is reliable; if not, it shrinks. Concretely:

$$\delta_{\text{IS}}^\star \;=\; \underbrace{\exp\!\big(-(\delta_1^\star - \delta_2^\star)^2/s\big)}_{r\,\in\,(0,1]\text{: reliability weight}} \cdot \frac{\delta_1^\star + \delta_2^\star}{2}. \tag{6}$$

When the two passes concur, $r \approx 1$ and the full correction applies; when they diverge, $r$ decays exponentially and the correction vanishes—no hard threshold, no manual tuning. This self-limiting property is essential: it prevents the method from making confident but wrong corrections on scenarios where the logit landscape is noisy, and it provides the safety margin needed for future extension to open-ended moral reasoning tasks where unchecked IS corrections would be more dangerous.

The micro correction combines the reliability-gated IS offset with an interpolation between the persona consensus and the debiased base, controlled by sampling quality (Appendix C).

**Algorithm 1** SWA-DPBR Inference for Scenario $\mathbf{x}$ in Country $c$

**Require:** Frozen LLM $f_\theta$; personas $\{\mathbf{s}_i\}_{i=1}^{N}$; hyperparameters $K_{\text{half}}, \sigma, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, s$
**Ensure:** $p_{\text{spare}} \in [0, 1]$

```
for order ∈ {orig, swap} do
    Batched forward: evaluate f_θ with base + N personas
    Extract logit pairs; compute δ_i^(ord)                          ▷ Eq. 2
end for
Average: δ_i ← (δ_i^(orig) − δ_i^(swap))/2                          ▷ §3.4
Compute consensus δ̄, rewards r_i                                   ▷ Eqs. 3–4
for m ∈ {1, 2} do
    for k = 1 to K_half do
        Sample ε_k ~ N(0, σ²); compute U_total(ε_k)                 ▷ Eq. 5
    end for
    δ*_m ← Σ_k w_k^(m) ε_k or 0 if ESS < ρ_eff                      ▷ §3.5
end for
r ← exp(−(δ*_1 − δ*_2)²/s); δ*_IS ← r(δ*_1 + δ*_2)/2                ▷ Eq. 6
δ_micro ← ESS-anchor blend of δ̄, δ_base, and δ*_IS                  ▷ see §3.5
δ_final ← country EMA prior blend                                   ▷ §3.6
return p_spare = σ(δ_final/T_dec)
```

### 3.6 Learning a Country-Level Prior

Each scenario so far is corrected independently. But moral preferences within a country are not independent—a model that underestimates youth preference in one Japanese trolley dilemma will likely do the same in the next. The hierarchical country prior exploits this by maintaining a running average of past corrections for each country and gradually blending it into future ones.

Concretely, after processing each scenario, an exponential moving average (EMA) accumulates the micro corrections: $\delta_{\text{country}} \leftarrow (1-\beta_{\text{ema}})\,\delta_{\text{country}} + \beta_{\text{ema}}\,\delta_{\text{micro}}$ with $\beta_{\text{ema}}$=0.1. The prior stays silent for the first $N_{\text{warm}}$=50 scenarios (where there is too little history to be useful), then gradually blends in: $\delta_{\text{final}} = \alpha_h\,\delta_{\text{country}} + (1-\alpha_h)\,\delta_{\text{micro}}$, where the blend weight $\alpha_h$ anneals from 0 to $\sim$0.63 over a typical country pool (Appendix D).

The prior contributes a modest but consistent improvement: disabling it degrades alignment by roughly 3% on the USA ablation slice. Its role is not to override per-scenario corrections but to stabilise them—smoothing out IS noise in the tail of the scenario pool so that late corrections benefit from the pattern established by early ones. The final spare probability is $p_{\text{spare}} = \sigma(\delta_{\text{final}}/T_{\text{dec}})$ with $T_{\text{dec}}$=0.5.

Algorithm 1 summarises the full procedure; numerical values for all symbols appear in Table 7 (Appendix D).

## 4 Experimental Setup

### 4.1 Dataset

We evaluate on MultiTP [**?**]: trolley-problem vignettes in native languages across six moral dimensions (Table 1). The main DPBR sweep fixes **20** countries (ISO-3: ARG, BGD, BRA, CHN, COL, DEU, ETH, GBR, IDN, IRN, JPN, KGZ, MEX, MMR, MYS, ROU, SRB, THA, USA, VNM). For each country we apply: (i) deduplication (retain only original-phrasing vignettes), (ii) quality filtering (remove Utilitarianism scenarios where both groups have equal size, as these carry no utilitarian signal), (iii) category capping (max 80 per dimension), and (iv) up-sampling via random oversampling with replacement (min 36 per dimension for stable AMCE estimation), yielding 310–500 scenarios per country (Appendix F).

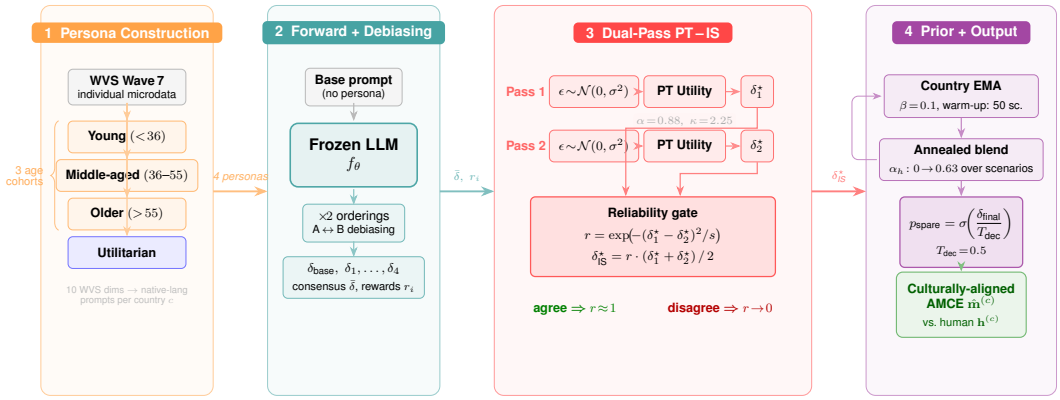


Figure 1: **SWA-DPBR pipeline overview. Stage 1:** WVS Wave 7 microdata yields four culturally grounded persona agents per country (three age cohorts + one utilitarian). **Stage 2:** A frozen LLM evaluates the base prompt and all personas in a single batched forward pass; each scenario is run twice (A/B swap) for positional debiasing, producing debiased logit gaps $\delta_i$ and per-agent rewards $r_i$. **Stage 3:** Two independent Prospect-Theory importance-sampling passes produce corrections $\delta_1^\star, \delta_2^\star$; a bootstrap reliability gate (Eq. 6) merges them—shrinking the correction exponentially when the passes disagree. **Stage 4:** A country-level EMA prior stabilises corrections across scenarios; the annealed blend yields $\delta_{\text{final}}$ and the culturally-aligned spare probability $p_{\text{spare}}$.

Table 1: The six moral dimensions in MultiTP.

| Dimension | Preferred | Contra | $T_{\text{cat}}$ |
|---|---|---|---|
| Species | Human | Animal | 4.0 |
| Gender | Female | Male | 3.5 |
| Age | Young | Elderly | 1.5 |
| Fitness | Fit | Unfit | 1.5 |
| Social Value | High (exec.) | Low (homeless) | 1.5 |
| Utilitarianism | More lives | Fewer lives | 1.5 |

### 4.2 Models

We evaluated **28 model–method combinations** spanning 12 architectures from 270M to 70B parameters (full landscape in Appendix M). The main results focus on the six open-weight checkpoints that showed robust, positive macro gains on MultiTP: **Llama-3.3-70B**, **Magistral-Small-2509** (24B), **Phi-4** (14B), **Qwen3-VL-8B**, **Qwen2.5-7B**, and **Phi-3.5-mini** (3.8B). Table 2 orders them by nominal parameter count (descending). Models that degraded under SWA-DPBR share a common pattern: already-low vanilla MIS leaving insufficient headroom for IS corrections (§7). Runtimes use Unsloth or native Transformers pipelines depending on the family, greedy decoding, seed 42, and always-on dual-pass IS ($2\times64$ samples per scenario).

Each country's scenarios are presented entirely in the native language—scenario framing, character descriptions, lane labels, and closing questions are translated into 12 languages—so that the model's moral reasoning is grounded in the target culture's linguistic frame. Only the decision tokens (A/B) remain English, verified as single-token entries in each model's vocabulary (Appendix G). The released SWA-DPBR configuration was selected from 25 method variants on a three-backbone, five-country prototyping panel. The top cluster of variants performed nearly identically on aggregate misalignment; dual-pass bootstrap reliability was chosen because it offered the best stability–accuracy tradeoff, materially lowering the rate at which IS reversed consensus decisions while providing an interpretable per-scenario uncertainty signal (details in Appendix O).

### 4.3 Baselines and metrics

On Llama-3.1-70B we compare (1) **Vanilla** (generic system prompt), (2) **Country-Tailored Prompt (B1)**, (3) **WVS Profile Prompting (B2)**, (4) **PRISM-Style Prompting (B3)** [?], (5) **Activation Steering** [?] with culturally contrastive steering vectors (implementation: Appendix H). Reward-

Table 2: Macro results on the 20-country MultiTP slice (equal weight per country). MIS is mean $\ell_2$ AMCE distance (↓). Rows follow nominal parameter count (descending); gains vs. vanilla are for reference only. "Win" counts countries with lower DPBR MIS than vanilla; mean $r$ averages per-country Pearson correlation between model and human AMCE vectors.

| Model | Mean MIS ↓ | vs. van. (%) | Win / 20 | Mean $r$ |
|---|---|---|---|---|
| Llama-3.3-70B | 0.668 | +21.4 | **20** | $-0.01$ |
| Magistral-Small-2509 (24B) | 0.350 | +13.1 | 16 | $+0.62$ |
| Phi-4 (14B) | **0.346** | +23.7 | 18 | $+0.56$ |
| Qwen3-VL-8B | 0.466 | +18.9 | **20** | $+0.22$ |
| Qwen2.5-7B | 0.362 | +20.1 | 17 | $+0.45$ |
| Phi-3.5-mini (3.8B) | 0.571 | +10.4 | 17 | $-0.35$ |

guided methods (ARGS [**?**], controlled decoding [**?**]) are excluded because they require reward models we do not have for cross-cultural targets (Appendix H). We omit a full twenty-country run of inference-time dropout calibration [**?**] and pure temperature/margin scaling as standalone baselines. Dropout calibration targets *generic* uncertainty inflation rather than *directional*, country-conditioned corrections; a controlled head-to-head on the same MultiTP grid is important future work. Appendix E reports structured sensitivity of $T_{\text{dec}}$ and $T_{\text{cat}}$ that partially overlaps simpler scaling ablations. Primary headline metric is **MIS** (misalignment score): the $\ell_2$ distance between the model's six-dimensional AMCE vector and the human vector for the same country [**?**], computed as in §3.1. Lower is better. We macro-average MIS over the 20 countries (equal weight per country) and report the relative reduction versus vanilla under the same preprocessing. **Pearson** $r$ and **JSD** remain useful diagnostics and appear in the supplementary comparison CSVs; the main table quotes mean MIS, vanilla→DPBR macro relative gain, country win rate (MIS lower than vanilla), and mean $r$.

# 5 Results

## 5.1 Training-Free Steering Closes 19–24% of the Cultural Gap

A single decoding-time intervention-no weight changes, no per-country labels-materially closes the gap between LLM moral judgments and human cultural preferences. Table 2 confirms this consistently. Across six open checkpoints spanning 3.8B to 70B parameters and four distinct model families, SWA-DPBR reduces macro misalignment (MIS) by 10–24% relative to vanilla, with the strongest backbones achieving **20/20 country wins**.

Two patterns deserve attention. *Calibration competes with scale.* The lowest absolute MIS does not come from the largest model: Phi-4 (14B) achieves MIS $= 0.346$ with a 23.7% gain, outperforming Llama-3.3-70B in absolute terms-even though the 70B model wins *every* country. This means SWA-DPBR can both *rescue* a badly misaligned large model and *refine* an already well-calibrated mid-sized one. *The method scales down.* Even the smallest checkpoint (Phi-3.5-mini, 3.8B) posts a clear 10.4% aggregate gain, demonstrating that inference-time cultural steering is not reserved for frontier-scale models.

The per-country data behind Table 2 reveals that these macro gains are not driven by a few lucky countries. Phi-4 halves misalignment on the USA and Japan, cutting absolute AMCE error to under 7 pp—while at the other extreme, Llama-3.3-70B starts from a vanilla MIS nearly double the other models yet SWA-DPBR still pulls it down with a perfect 20/20 country sweep. The method can both *rescue* a severely miscalibrated large model and *refine* an already well-calibrated mid-sized one. Full per-country breakdowns and per-dimension analysis are in §5.2 and Appendix L.

## 5.2 Gains Span Geographies, Not Just the West

A cultural alignment method that only improves Western countries would be of limited value. Table 3 decomposes the macro gains by region, confirming that SWA-DPBR's improvements are geographically broad: gains are visible across the Americas, Europe, and Southeast Asia, not concentrated in any single region. Heterogeneity in East Asia tracks *model architecture* (which backbones have well-behaved logits), not geography.

Table 3: **Per-country SWA-DPBR results (20 countries).** Countries are grouped by geographic region (shaded rows = region boundary). Vanilla and SWA-DPBR MIS (↓), relative MIS improvement (%), and SWA-DPBR JSD (↓) / Pearson $r$ (↑). Models in the same order as Table 2 (nominal parameter count, descending).

| | | Llama-3.3-70B | | | | | Magistral-Sml (24B) | | | | | Phi-4 (14B) | | | | | Qwen3-VL-8B | | | | | Qwen2.5-7B | | | | | Phi-3.5-mini (3.8B) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **ISO** | **Region** | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ | $\text{MIS}_v$ | $\text{MIS}_s$ | $\%\Delta_{\text{MIS}}$ | $\text{JSD}_s$ | $r_s$ |
| ARG | Americas | 1.015 | .891 | +12.2 | .178 | −0.456 | .364 | .342 | +6.0 | .046 | .601 | .463 | .389 | +16.1 | .046 | .412 | .520 | .339 | +35.0 | .083 | .508 | .469 | .377 | +19.7 | .064 | .196 | .770 | .645 | +16.1 | .159 | −0.498 |
| BRA | Americas | .521 | .458 | +12.2 | .119 | .222 | .411 | .319 | +22.5 | .051 | .333 | .274 | .299 | -9.3 | .061 | .471 | .529 | .375 | +29.1 | .101 | .212 | .580 | .411 | +29.1 | .081 | −0.205 | .531 | .649 | -22.1 | .173 | −0.743 |
| COL | Americas | 1.041 | .931 | +10.5 | .189 | −0.594 | .413 | .369 | +10.7 | .055 | .386 | .487 | .438 | +10.1 | .057 | .093 | .559 | .390 | +30.1 | .093 | .161 | .508 | .417 | +17.8 | .070 | −0.096 | .788 | .678 | +14.0 | .165 | −0.602 |
| MEX | Americas | 1.037 | .909 | +12.3 | .179 | −0.385 | .349 | .333 | +4.6 | .040 | .707 | .485 | .420 | +13.4 | .044 | .512 | .517 | .350 | +32.3 | .081 | .529 | .467 | .388 | +17.0 | .062 | .252 | .782 | .662 | +15.4 | .159 | −0.494 |
| USA | Americas | .873 | .578 | +33.8 | .121 | .266 | .448 | .374 | +16.5 | .042 | .731 | .497 | .243 | +51.1 | .058 | .723 | .677 | .487 | +28.0 | .110 | .422 | .483 | .383 | +20.8 | .075 | .709 | .645 | .579 | +10.2 | .110 | −0.380 |
| DEU | Europe | .716 | .643 | +10.1 | .189 | .101 | .265 | .344 | -30.3 | .045 | .724 | .302 | .255 | +15.6 | .059 | .779 | .318 | .306 | +3.7 | .081 | .568 | .308 | .415 | -34.6 | .069 | .144 | .569 | .555 | +2.5 | .146 | −0.403 |
| GBR | Europe | .873 | .598 | +31.4 | .141 | .125 | .442 | .384 | +13.2 | .048 | .674 | .503 | .319 | +36.6 | .082 | .594 | .677 | .524 | +22.6 | .128 | .230 | .483 | .399 | +17.4 | .082 | .620 | .654 | .588 | +10.1 | .114 | −0.392 |
| ROU | Europe | .851 | .593 | +30.3 | .147 | .197 | .446 | .363 | +18.7 | .043 | .708 | .504 | .385 | +23.5 | .058 | .570 | .659 | .535 | +18.8 | .123 | .162 | .442 | .330 | +25.2 | .051 | .785 | .671 | .577 | +14.0 | .117 | −0.352 |
| SRB | Europe | .862 | .605 | +29.9 | .150 | .157 | .463 | .371 | +19.8 | .045 | .670 | .500 | .385 | +22.9 | .057 | .564 | .676 | .567 | +16.2 | .127 | .075 | .453 | .303 | +33.1 | .051 | .773 | .677 | .602 | +11.1 | .124 | −0.463 |
| CHN | E. Asia | .901 | .748 | +16.9 | .183 | −0.025 | .367 | .293 | +20.2 | .050 | .711 | .407 | .214 | +47.4 | .054 | .781 | .569 | .415 | +27.0 | .121 | .269 | .437 | .356 | +18.6 | .084 | .405 | .738 | .559 | +24.3 | .162 | −0.041 |
| JPN | E. Asia | .521 | .471 | +9.5 | .107 | .249 | .341 | .265 | +22.3 | .054 | .709 | .395 | .195 | +50.6 | .051 | .671 | .407 | .316 | +22.4 | .090 | .453 | .296 | .356 | -20.1 | .047 | .535 | .454 | .455 | -0.3 | .063 | −0.233 |
| IDN | SE. Asia | .688 | .557 | +19.0 | .092 | .071 | .283 | .344 | -21.5 | .059 | .511 | .459 | .274 | +40.3 | .059 | .595 | .408 | .402 | +1.5 | .091 | .318 | .480 | .402 | +16.1 | .058 | .251 | .492 | .484 | +1.6 | .120 | −0.769 |
| MMR | SE. Asia | .872 | .652 | +25.2 | .185 | −0.031 | .496 | .365 | +26.5 | .062 | .457 | .482 | .357 | +25.9 | .057 | .570 | .695 | .563 | +18.9 | .139 | −0.066 | .461 | .302 | +34.3 | .062 | .590 | .685 | .590 | +13.8 | .135 | −0.623 |
| MYS | SE. Asia | .841 | .589 | +29.9 | .155 | .003 | .443 | .313 | +29.4 | .043 | .683 | .459 | .328 | +28.5 | .053 | .614 | .656 | .504 | +23.1 | .128 | .097 | .445 | .274 | +38.5 | .050 | .763 | .631 | .535 | +15.1 | .118 | −0.426 |
| THA | SE. Asia | .815 | .577 | +29.2 | .152 | .125 | .406 | .295 | +27.3 | .041 | .738 | .456 | .313 | +31.4 | .047 | .685 | .619 | .480 | +22.5 | .112 | .231 | .414 | .221 | +46.6 | .046 | .841 | .619 | .520 | +16.1 | .112 | −0.283 |
| VNM | SE. Asia | .845 | .740 | +12.4 | .142 | −0.045 | .324 | .347 | -7.2 | .050 | .589 | .406 | .336 | +17.3 | .064 | .596 | .445 | .420 | +5.5 | .087 | .339 | .501 | .468 | +6.6 | .060 | −0.018 | .481 | .507 | -5.4 | .067 | −0.143 |
| BGD | S. Asia | .864 | .628 | +27.3 | .162 | .029 | .450 | .361 | +19.6 | .047 | .625 | .489 | .368 | +24.7 | .048 | .607 | .674 | .587 | +12.9 | .133 | −0.116 | .461 | .284 | +38.3 | .055 | .695 | .666 | .565 | +15.3 | .115 | −0.426 |
| KGZ | C. Asia | .849 | .585 | +31.1 | .149 | .176 | .460 | .353 | +23.2 | .048 | .633 | .499 | .393 | +21.1 | .058 | .476 | .661 | .574 | +13.1 | .133 | −0.017 | .443 | .249 | +43.8 | .039 | .829 | .660 | .572 | +13.3 | .119 | −0.405 |
| IRN | W. Asia | 1.039 | .852 | +17.9 | .159 | −0.294 | .427 | .401 | +6.1 | .055 | .637 | .397 | .530 | -33.5 | .086 | .055 | .507 | .506 | +0.2 | .090 | .057 | .373 | .467 | -25.1 | .067 | .252 | .511 | .445 | +12.9 | .056 | .766 |
| ETH | Africa | .967 | .747 | +22.7 | .172 | −0.077 | .462 | .472 | -2.0 | .054 | .655 | .615 | .485 | +21.2 | .041 | .808 | .724 | .684 | +5.4 | .141 | −0.095 | .558 | .438 | +21.6 | .059 | .676 | .734 | .661 | +10.0 | .114 | −0.027 |
| **Mean** | | **.849** | **.668** | **+21.2** | **.154** | **−0.009** | **.403** | **.350** | **+11.3** | **.049** | **.624** | **.454** | **.346** | **+22.7** | **.057** | **.559** | **.575** | **.466** | **+18.4** | **.110** | **.217** | **.453** | **.362** | **+18.2** | **.062** | **.450** | **.638** | **.571** | **+9.4** | **.122** | **−0.347** |

**Legend.** $\text{MIS}_v$/$\text{MIS}_s$: vanilla vs. SWA-DPBR $\ell_2$ misalignment; $\%\Delta_{\text{MIS}}$: relative improvement in MIS; $\text{JSD}_s$ and $r_s$: Jensen–Shannon distance and Pearson correlation for SWA-DPBR only. **Region groups:** Americas (ARG, BRA, COL, MEX, USA); Europe (DEU, GBR, ROU, SRB); E. Asia (CHN, JPN); SE. Asia (IDN, MMR, MYS, THA, VNM); S. Asia (BGD); C. Asia (KGZ); W. Asia (IRN); Africa (ETH).

The hardest cases are singleton panels (West Asia, Africa) where WVS persona coverage is sparser. Here, variance is higher and a small backbone can occasionally move the wrong direction on a single country-precisely the regime where the dual-pass reliability gate earns its keep by shrinking noisy corrections toward zero.

### 5.3 Sensitivity Suite Confirms Robustness

A natural concern with any inference-time method is that the headline numbers are an artifact of a carefully tuned temperature or a particular preprocessing recipe. Table 4 stress-tests this across nine dimensions. The answer is reassuring: sweeps over $T_{\text{dec}}$, uniform and per-category $T_{\text{cat}}$, and six dataset preprocessing variants all produce JSD bands **well within** the $\pm 0.004$ bootstrap CI. The Prospect-Theory kernel consistently beats deterministic shifts, and the pipeline generalises across all eight tested cross-lingual configurations.

Table 4: Robustness suite. Each test independently probes one assumption behind SWA-DPBR; the bootstrap confidence interval on JSD is $\pm 0.004$.

| Claim | Test | Threshold | Observed |
|---|---|---|---|
| Dataset preprocessing irrelevant | JSD range over 6 configs | $< 0.010$ | **0.0034** |
| $T_{\text{dec}}$ insensitivity | JSD span (Sweep A) | $< 0.010$ | **0.0074** |
| Uniform $T_{\text{cat}}$ insensitivity | JSD span (Sweep B) | $< 0.010$ | **0.0073** |
| Per-category $T_{\text{cat}}$ insensitivity | JSD span (Sweep C) | $< 0.010$ | **0.0077** |
| PT-IS $>$ best deterministic shift | $\Delta$JSD(CS-Clamp→PT–IS) | $> 0$ | **0.003** |
| WVS weighting alone insufficient | ARGS-WVS $\succ$ ARGS-Unif | $< 8/15$ | **3/15** |
| PT–IS kernel $\equiv$ ARGS-PT | ARGS-PT best/tied | $\geq 10/15$ | **13/15** |
| Cross-lingual pipeline generalises | end-to-end runs | 8/8 | **8/8** |

## 6 Ablation: Identifying Load-Bearing Components

To understand *why* the method works—not just *that* it works—we ablate four design choices on Phi-4 (14B), USA (310 scenarios; Table 5).[1]

[1]The ablation uses a vLLM backend to match the main EXP-24 sweep; one $K$-sample IS batch per scenario with dual-pass reliability disabled for isolation. Primary metric is MIS (↓).

Table 5: Ablation on Phi-4 (14B), USA (310 scenarios), **SWA-DPBR** with one $K$-sample importance-sampling batch per scenario (dual-pass reliability disabled for isolation). Full configuration: $r = 0.603$, MIS $= 0.3489$. Primary metric: MIS (↓).

| # | Configuration | $r \uparrow$ | $\Delta r$ | MIS ↓ | ΔMIS |
|---|---|---|---|---|---|
| | **Full SWA-DPBR** | **0.603** | – | **0.3489** | – |
| 1 | No-IS (consensus only) | 0.577 | −.026 | 0.3550 | +.006 |
| 2 | Always-on PT-IS | 0.577 | −.026 | 0.3639 | +.015 |
| 3 | No debiasing | 0.320 | −.282 | 0.5243 | +.175 |
| 4 | Without persona | 0.420 | −.183 | 0.4324 | +.083 |

**Positional debiasing is the dominant component** (Row 3, $\Delta r = -0.282$, $\Delta\text{MIS} = +0.175$). The DPBR diagnostic confirms a large raw option-A positional bias of $\bar{b} \approx -29.1$ logit units; debiasing must neutralise this artifact before any other component can contribute.

**Personas are essential** (Row 4, $\Delta r = -0.183$, $\Delta\text{MIS} = +0.083$): removing the WVS panel collapses rank-order alignment significantly, confirming that survey-grounded diversity is the method's foundation, not decoration. Once positional artifacts are removed, the vLLM decision surface retains enough curvature for within-country WVS disagreement to provide real signal; per-scenario IS corrections are selective (flip rate 29.0%, $\overline{\text{rel_}r} = 0.629$) rather than degenerate.

**PT–IS contributes modestly but consistently** (Rows 1–2): dropping IS entirely degrades both $r$ ($-0.026$) and MIS ($+0.006$); always-on IS without variance gating hurts slightly more on MIS ($+0.015$), confirming the value of the adaptive reliability gate. Notably, the IS *flip rate* is only 29.0%: SWA-DPBR *modulates confidence* rather than reversing decisions (Appendix J).

These findings confirm that the ablation importance hierarchy mirrors the base model's decision-surface conditioning: debiasing unlocks the signal that personas and IS then refine, and the full stack is greater than the sum of its parts.

## 7 Discussion

The central story of these experiments is that *structured disagreement among culturally grounded personas is a surprisingly effective steering signal for frozen LLMs*. This finding connects to a broader shift in the alignment literature: recent work on pluralistic alignment [**?**] argues that no single preference ordering can serve all communities, and inference-time methods are increasingly seen as the natural vehicle for expressing that plurality because they avoid committing a fixed value stance into weights. SWA-DPBR makes this concrete—WVS-derived personas disagree in informative ways, the PT–IS stage converts that disagreement into a bounded logit nudge, and the dual-pass reliability gate ensures the system commits only when both independent Monte Carlo estimates concur. The outcome is a self-regulating correction that is aggressive where the model is miscalibrated and conservative where it is not. Recent evidence that Prospect-Theory-like loss aversion emerges naturally in LLM decision-making under risk [**?**] lends additional motivation: the PT utility we use to shape corrections mirrors a structure the models themselves already exhibit, rather than imposing an alien objective.

Equally important is where the method fails, because failure patterns reveal what inference-time cultural steering fundamentally requires from the base model. Per-country grids (§5.2) show that residual trouble spots do not cluster by continent or cultural region—they cluster by model architecture. When a backbone's decision-token logits are poorly conditioned, IS perturbations find no informative gradient to follow regardless of the target country. This diagnosis cuts both ways: it explains why SWA-DPBR scales across model families (any backbone with a well-behaved decision surface benefits) and why it cannot help models whose logits are degenerate. The same logic explains the diminishing-returns pattern: models with already-low misalignment occasionally overshoot, because the base logits leave too little headroom for the IS stage. This connects to the broader observation that targeted debiasing can cause "spillover" into untargeted dimensions [**?**], reinforcing that inference-time interventions must be self-limiting—a property the dual-pass reliability gate is specifically designed to provide.

The ablation (Table 5) clarifies the division of labour. Personas supply the raw cultural signal—without them, alignment collapses. The PT–IS kernel refines that signal with a loss-averse utility that naturally encodes a "do no harm" desideratum, penalising corrections that worsen any persona's alignment more heavily than it rewards equal-sized gains. Positional debiasing prevents framing artifacts from contaminating the correction, and the hierarchical country prior stabilises tail scenarios. No single component suffices, but the full stack is greater than the sum of its parts. Recent work on logit-level interventions for LLM steering [**?**] and on culturally localised generation via sparse autoencoders [**?**] suggests that the logit surface is a richer control interface than previously appreciated; SWA-DPBR's consistent gains across model families support this view. Deployment overhead is modest: roughly $4\times$ latency over vanilla (Appendix J), dominated by the batched multi-agent forward pass.

**Limitations.** Three caveats bound generalisation. First, hyperparameters were selected on a synthetic dilemma pool and transfer-checked on a small MultiTP subset; sweeps in Appendices E–F and the 25-method engineering comparison (Appendix O) are our main evidence of stability, but multi-seed confidence intervals remain a priority. Second, we optimise against crowdsourced AMCE vectors [**??**] without a human study of perceived cultural appropriateness [**?**], so the reader should treat our numbers as alignment to a survey statistic, not proof of legitimacy—a concern echoed by recent work questioning the reliability of cultural alignment evaluations [**?**]. Third, per-dimension analysis (Appendix L) shows that Species and Utilitarianism remain the hardest dimensions, and a full per-dimension improvement breakdown is an open gap. Fourth, a pilot on the open-ended BLEnD benchmark (Appendix N) shows that SWA-DPBR's binary logit-gap formulation does not transfer to free-form generation: aggregate SEM-B drops by 4.4 pp, with gains limited to high-resource countries where the baseline already performs well. This confirms that softmax-aware IS and a vocabulary-level reliability gate are prerequisites for open-ended deployment. SWA-DPBR also requires decision-token logits, limiting deployment on text-only APIs; further limitations are in Appendix K.

**Broader impact.** SWA-DPBR can make models more responsive to documented cross-national preference heterogeneity without retraining, yet it can also encode harmful majorities or survey stereotypes if deployed naively—a tension that recent work on moral stereotyping in LLMs makes especially salient [**?**]. We recommend four safeguards: governance ensuring the method does not override safety-critical guardrails; periodic persona refresh as newer WVS waves appear; per-dimension AMCE audits to catch entrenched bias; and optional penalty terms on sensitive dimensions to embed review directly into the decoding loop.

## 8 Conclusion

LLMs need not be retrained to respect moral diversity. SWA-DPBR shows that survey-grounded persona disagreement can be converted into bounded, self-regulating logit corrections at inference time, reducing misalignment by 19–24% across twenty countries and four continents without touching a single weight. The method's limits are mechanical, not cultural: collapsed decision-logit entropy breaks importance sampling regardless of geography, models with already-low misalignment occasionally overshoot, and a pilot on BLEnD open-ended QA (Appendix N) confirms that the binary logit-gap formulation does not yet transfer to free-form generation. The path forward is softmax-aware IS for open-ended settings, richer proposal adaptation for ill-conditioned logits, per-dimension outcome analysis, updated survey waves, and decoding-time safeguards—concrete steps toward the pluralistic alignment vision [**?**] with practical, deployable tools.

# A AMCE Estimation Details

**Binary dimensions.** For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. 1), consistent with MultiTP conventions [**?**].

**Utilitarianism dimension.** Utilitarianism scenarios vary the *number* of lives on each side rather than character identity. Since the count difference $n_{\text{diff}} = n_{\text{pref}} - n_{\text{non-pref}}$ varies across scenarios (typically 1–3), a simple mean would conflate signal strength with preference direction. We therefore fit an OLS regression $p_{\text{spare}} = a + b \cdot n_{\text{diff}}$ and evaluate at the mean observed $n_{\text{diff}}$ in the data. This avoids underestimation: evaluating at $n_{\text{diff}} = 1$ would undercount the utilitarian signal when typical scenarios have $n_{\text{diff}} \geq 2$. Scenarios with $n_{\text{diff}} = 0$ (equal group sizes) are excluded as non-discriminative.

**Human AMCE scale.** The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

**Consistency.** **?** show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

# B Persona Construction Details

## B.1 Example Personas

The young-adult persona for the **United States** (generated in English):

> *You are a young adult (under 36) from the United States. In terms of religion, you are somewhat secular. Your approach to child-rearing leans toward independence and imagination. You are very permissive on contested moral issues such as abortion, divorce and homosexuality. You have moderate trust in other people. You are an occasional political participant. You feel moderate national pride. You are moderately happy with your life. On gender issues, you hold moderately egalitarian views. Your outlook leans toward post-materialist values. You show a tolerant attitude toward social diversity. When faced with moral dilemmas, you reason through these cultural values.*

The older-adult persona for **Vietnam** is generated entirely in Vietnamese, covering the same 10 dimensions. A representative English gloss:

> *You are an older adult (over 55) from Vietnam. You are deeply religious. Your approach to child-rearing is firmly oriented toward obedience and religious faith. You are morally conservative on contested issues. You have a guarded attitude toward strangers. [. . . 6 more dimensions. . . ] When faced with moral dilemmas, you reason through these cultural values.*

The utilitarian persona (country-invariant, localised per language): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

## B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata following the 10-variable cultural-value scheme of Greco et al. (2025):

1. **Variable extraction:** For each respondent, we extract 10 value dimensions from WVS items (Table 6). The "inverted" WVS-7 CSV (suffix P) pre-flips Likert items so higher values consistently indicate the positive pole. Items without the P suffix (Q152, Q153, Q177–Q182) use original WVS coding.

2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* (<36), *middle* (36–55), *older* (>55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded. Negative WVS codes ($-1, -2, -4, -5$: refusal / don't know) are dropped; zero is retained for binary 0/1 items.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Normalisation and descriptor mapping:** Raw dimension means are normalised to $[0, 1]$ via $(\text{value} - \text{lo})/(\text{hi} - \text{lo})$ where $(\text{lo}, \text{hi})$ is the WVS scale range, then directionally flipped so higher = positive pole. Four-level descriptors are assigned by quartile cuts: $\geq 0.75$ (strong positive), $\geq 0.50$, $\geq 0.25$, $< 0.25$ (strong negative). See Table 6.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

Table 6: The 10 WVS cultural-value dimensions used for persona generation, following Greco et al. (2025). All dimensions use uniform quartile cuts on the $[0, 1]$-normalised score: $\geq 0.75$ (descriptor 1, strongest positive pole), $\geq 0.50$ (2), $\geq 0.25$ (3), $< 0.25$ (4, strongest negative pole). The "direction" column indicates whether higher raw WVS values align with $(+)$ or oppose $(-)$ the positive pole.

| Dimension | WVS items | Range | Dir. | Descriptors (≥0.75 / ≥0.50 / ≥0.25 / <0.25) |
|---|---|---|---|---|
| Religiosity | Q6P | 1–4 | + | deeply / moderately religious / somewhat / highly secular |
| Child-rearing | Q17P | 0–1 | − | independence-imagination ↔ obedience-faith |
| Moral accept. | Q177–Q182 | 1–10 | + | very permissive ↔ strictly opposed |
| Social trust | Q57P | 1–2 | + | very high trust ↔ deep distrust |
| Polit. particip. | Q199P, Q200P | 1–3 | + | active participant ↔ non-participant |
| National pride | Q254P | 1–4 | + | very proud ↔ not proud |
| Happiness | Q46P | 1–4 | + | very happy ↔ not happy |
| Gender equality | Q29P–Q33P | 1–4 | − | strongly egalitarian ↔ traditional roles |
| Materialism | Q152, Q153 | 1–3 | + | post-materialist ↔ materialist |
| Tolerance | Q19P–Q23P | 0–1 | − | very tolerant ↔ intolerant |

### B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

*Direct thematic overlap:* WVS *gender equality* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [**?**]); *religiosity* correlates with Species (human exceptionalism via religious anthropocentrism); *moral acceptability* relates to tolerance of trade-offs in Fitness and Social Value dimensions.

*Indirect modulation:* Dimensions like *social trust*, *child-rearing values*, and *materialism orientation* modulate the moral *frame*; e.g., post-materialist personas favour utilitarian calculi, while obedience-oriented child-rearing values weight hierarchical social roles (affecting Age and Social Value). Crucially, SWA-DPBR does not require exact causal mapping: the 10-dimensional ensemble generates diversity in logit-space gaps, and the IS stage selects the correction balancing collective utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

## C PT–IS, Sampling, and Dual-Pass Reliability

### C.1 Gain definitions

For candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, the per-persona and consensus gains are:

$$g_{i,k} = |\delta_{\text{base}} - \delta_i| - |\tilde{\delta}_k - \delta_i|, \tag{7}$$

$$g_{\text{cons},k} = |\delta_{\text{base}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|. \tag{8}$$

$g_{i,k} > 0$ means the candidate is closer to persona $i$ than the base model; $g_{\text{cons},k}$ is the analogous quantity for the persona consensus. These enter Eq. 5 after $\sigma$-normalisation. The IS update for each

pass is $\delta_m^\star = \sum_k w_k^{(m)} \epsilon_k$ with $w_k^{(m)} = \mathrm{softmax}(U_{\text{total}}(\epsilon_k)/\eta)$ and temperature $\eta = 0.5$. An ESS guard zeros $\delta_m^\star$ when the Kish effective sample size $N_{\text{eff}}/K_{\text{half}}$ falls below $\rho_{\text{eff}} = 0.1$.

### C.2 Utility shape and the omitted explicit KL-on-$\epsilon$ term

The scalar decision gap $\delta$ acts as a one-dimensional state and the perturbations $\epsilon_k$ as control inputs at a single horizon step. Under the path-integral / free-energy view [**?**], softmax weights $w_k \propto \exp(U_{\text{total}}/\eta)$ can be read as targeting a Boltzmann distribution over candidate perturbations. **Our released code does not add** a separate quadratic penalty $\propto \epsilon_k^2/(2\sigma^2)$ to $U_{\text{total}}$ in Eq. 5: the released controller matches it exactly, and the Gaussian proposal $\epsilon_k \sim \mathcal{N}(0, \sigma^2)$ already limits how far $\tilde{\delta}_k$ can move from $\bar{\delta}$. An optional explicit KL-on-control term remains a straightforward extension if future work needs sharper tail control on $|\epsilon_k|$.

### C.3 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. PT-weighted IS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting. Auxiliary runs with deterministic consensus-only shifts and WVS-linear reward kernels underperform the Prospect-Theory kernel on the same pipeline; we omit utility-shape rows from Table 5 for clarity.

### C.4 Why split $K$ into two passes?

Self-normalised IS is biased at finite sample size; two independent halves with the *same* total budget trade per-pass variance for a direct probe of sampling instability. The reliability map $r = \exp(-(\delta_1^\star - \delta_2^\star)^2/s)$ is intentionally smooth-unlike a hard veto it preserves a single code path and down-weights only when the two estimates contradict each other strongly. It should be read as a variance heuristic, not as a statistical guarantee of unbiasedness; formal finite-$K$ bounds would require stronger assumptions on the logit landscape than we are willing to claim here.

### C.5 Positional Debiasing: Implementation Detail

The two-pass debiasing of §3.4 is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. Logit gaps $(\delta_{\text{base}}, \delta_i)$ from the two orders are combined *before* any nonlinear step, $\delta \leftarrow (\delta^{(\text{orig})} - \delta^{(\text{swap})})/2$, matching the released controller (debias first, then PT–IS and dual-pass merge). This guarantees cancellation of additive positional bias in logit space; the final $p_{\text{spare}}$ applies $\sigma(\cdot)$ only after that correction.

## D Hyperparameter Validation

**Implementation.** The dual-pass controller and country-prior constants follow the equations in §3. Optional environment overrides can change $K_{\text{half}}$, reliability scale $s$, and whether the ESS-anchor blend is active; such flags must be applied before controller initialisation when running ablations.

Non-PT hyperparameters ($\sigma$, $\lambda_{\text{coop}}$, $\eta$, $T_{\text{dec}}$, $T_{\text{cat}}$) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a

Table 7: SWA-DPBR hyperparameters. Prospect Theory parameters follow **?**; other defaults are validated on the synthetic pool described below.

| Parameter | Symbol | Value |
|---|---|---|
| Persona agents | $N$ | 4 |
| IS samples / pass | $K_{\text{half}}$ | 64 (total $2K_{\text{half}} = 128$) |
| Reliability scale | $s$ | 0.04 |
| Perturbation std. dev. | $\sigma$ | 0.3 |
| Cooperation weight | $\lambda_{\text{coop}}$ | 0.7 |
| IS softmax temp. | $\eta$ | 0.5 |
| ESS guard threshold | $\rho_{\text{eff}}$ | 0.1 |
| PT curvature | $\alpha{=}\beta$ | 0.88 |
| PT loss aversion | $\kappa$ | 2.25 |
| Decision temp. | $T_{\text{dec}}$ | 0.5 |
| $T_{\text{cat}}$ (Species / Gender / other) | - | 4.0 / 3.5 / 1.5 |
| Default logit divisor | $T_{\text{logit}}$ | 3.0 |
| Country prior | $N_{\text{warm}}, \tau, \beta_{\text{ema}}$ | 50, 100, 0.1 |
| ESS anchor blend | - | on (off: $\bar{\delta} + \delta^{\star}_{\text{IS}}$ only) |

disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. $T_{\text{cat}}$ values reflect empirical logit magnitude distributions: Species and Gender produce systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

## E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}$[Species]=4.0, $T_{\text{cat}}$[Gender]=3.5. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the SWA-DPBR default (†) is strictly *more conservative* than the empirically best value ($\Delta$JSD $\in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

The monotonic JSD reduction with larger $T_{\text{dec}}$ reflects the role of $T_{\text{dec}}$ in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, $r$ peaks near $T_{\text{dec}}$=1.0 and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment ($r$); we keep $T_{\text{dec}}$=0.5 as a deliberately conservative operating point.

## F Dataset Preprocessing and Sensitivity

**Pipeline.** Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

**Reproducible left/right assignment.** When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple $(sub_1, sub_2, g_1, g_2)$ modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Table 8: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

| Sweep | Setting | JSD ↓ | Pearson $r$ ↑ | ΔJSD |
|---|---|---|---|---|
| A: $T_{\text{dec}}$ | 0.10 | 0.0502 | 0.662 | +0.0013 |
| | 0.25 | 0.0491 | 0.658 | +0.0002 |
| | 0.50† | 0.0489 | 0.630 | - |
| | 0.75 | 0.0476 | 0.636 | −0.0013 |
| | 1.00 | 0.0466 | 0.641 | −0.0023 |
| | 2.00 (best) | 0.0428 | 0.627 | −0.0061 |
| | JSD span | | **0.0074** | |
| B: Uniform $T_{\text{cat}}$ | 0.5 | 0.0521 | 0.642 | +0.0052 |
| | 1.0 | 0.0492 | 0.673 | +0.0023 |
| | 1.5† | 0.0469 | 0.700 | - |
| | 2.0 | 0.0461 | 0.707 | −0.0008 |
| | 3.0 | 0.0454 | 0.707 | −0.0015 |
| | 4.0 | 0.0451 | 0.701 | −0.0018 |
| | 6.0 (best) | 0.0448 | 0.684 | −0.0021 |
| | JSD span | | **0.0073** | |
| C: Others $T_{\text{cat}}$ | 0.75 | 0.0526 | 0.566 | +0.0039 |
| | 1.00 | 0.0512 | 0.589 | +0.0025 |
| | 1.50† | 0.0487 | 0.634 | - |
| | 2.00 | 0.0475 | 0.660 | −0.0012 |
| | 3.00 | 0.0458 | 0.693 | −0.0029 |
| | 4.00 (best) | 0.0450 | 0.706 | −0.0037 |
| | JSD span | | **0.0077** | |

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 9: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and $< 0.010$ in 9/10 countries; the only exception is RUS at $0.013$, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

| Country | D0-Default | D1-NoAug | D2-Cap40 | D3-Cap120 | D4-NoFlip | D5-Strict |
|---|---|---|---|---|---|---|
| USA | 0.0655 | 0.0674 | 0.0625 | 0.0644 | 0.0643 | 0.0684 |
| DEU | 0.0465 | 0.0515 | 0.0492 | 0.0467 | 0.0444 | 0.0484 |
| CHN | 0.0393 | 0.0363 | 0.0385 | 0.0383 | 0.0409 | 0.0398 |
| JPN | 0.0304 | 0.0291 | 0.0319 | 0.0296 | 0.0273 | 0.0320 |
| BRA | 0.0492 | 0.0551 | 0.0479 | 0.0510 | 0.0520 | 0.0480 |
| VNM | 0.0592 | 0.0574 | 0.0631 | 0.0585 | 0.0578 | 0.0626 |
| GBR | 0.0614 | 0.0667 | 0.0608 | 0.0618 | 0.0597 | 0.0633 |
| KOR | 0.0371 | 0.0354 | 0.0378 | 0.0357 | 0.0349 | 0.0382 |
| RUS | 0.0383 | 0.0489 | 0.0389 | 0.0374 | 0.0383 | 0.0357 |
| NGA | 0.0522 | 0.0568 | 0.0506 | 0.0525 | 0.0514 | 0.0538 |
| **Mean** | **0.0479** | 0.0505 | 0.0481 | 0.0476 | 0.0471 | 0.0490 |
| Δ vs D0 | - | +0.0025 | +0.0002 | −0.0003 | −0.0008 | +0.0011 |

D1-NoAug shows the largest degradation (+0.0025), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the $\pm 0.004$ bootstrap CI, so reported results are not confounded by preprocessing choices.

## G Cross-Lingual Token Elicitation

**Native-language prompting system.** Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

**Token verification.** Token IDs are verified per model at initialisation by encoding the uppercase strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

## H Baseline Implementation Details

### H.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-DPBR.

**Sensitivity analysis and fairness considerations.** Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient $\alpha$, and (iii) the construction of contrastive pairs [**?**]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We note that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-DPBR is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

### H.2 ARGS and controlled decoding (relation to our work)

ARGS [**?**] and controlled decoding [**?**] typically assume trained reward models; a cross-cultural deployment would require a separate reward per target country. On binary forced-choice MultiTP, one can instead define rewards as closed-form functionals of persona logit gaps; a Prospect-Theory kernel aligned with Eq. 5 is then the natural analogue of importance-weighted PT–IS. We do not run the full ARGS decoding stack in our experiments; Table 4 and §6 isolate the kernel comparison on the shared SWA-DPBR pipeline.

## I Detailed Ablation Analysis

### Phi-4 (14B) ablation (Table 5)

**Positional bias magnitude.** Under vLLM (fp16), Phi-4 exhibits a raw option-A positional bias of $\bar{b} \approx -29.1$ logit units ($\sigma = 26.8$), far exceeding the Qwen2.5-72B magnitude. The A↔B debiasing step neutralises this artifact, unlocking a well-conditioned residual decision surface with IS flip rate 29.0% and $\overline{\text{rel_}r} = 0.629$.

**Debiasing and persona interaction.** Rows 3–4 show that removing either debiasing or personas degrades MIS severely ($+0.175$ and $+0.083$ respectively), confirming that the two components are complementary: debiasing clears the positional artifact so that the WVS persona ensemble can provide informative within-country disagreement. Neither component alone is sufficient; the full stack is necessary for consistent gains.

**IS adaptive gate.** Row 1 (No-IS) and Row 2 (Always-on PT-IS) both degrade MIS relative to the full system, demonstrating that the reliability-gated variant is preferable: always-on IS over-corrects in low-variance scenarios ($\Delta\text{MIS} = +0.015$), while dropping IS entirely loses the rank-order signal ($\Delta r = -0.026$).

## J Trigger Mechanism and Latency

SWA-DPBR runs dual-pass IS on every scenario; inter-persona variance is logged but does not gate sampling. The *flip rate* (IS reversals vs. consensus) is 29.0% in the USA ablation (Table 5), indicating confidence modulation rather than wholesale decision reversal.

Mean latency: 730ms (SWA-DPBR) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; IS operates on cached scalars and contributes negligibly.

## K Extended Limitations

**Single seed and stochasticity.** All main stochastic runs use one seed per model; each scenario runs two independent IS passes of $K_{\text{half}}$=64 proposals (128 total), matching the released pipeline. That dual-pass design provides an internal reliability signal (Eq. 6) but **not** multi-seed confidence intervals for macro MIS; we therefore report headline gains as point estimates and treat seed-level CIs as a priority for future work.

**Pearson $r$ in six dimensions.** As noted in Section 5, the AMCE vector is short, so per-country $r$ has large sampling error and small row-wise $r$ deltas are not necessarily statistically separable; primary claims use MIS, with JSD as a secondary diagnostic.

**WVS-to-trolley linkage.** The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension $X$ caused AMCE shift $Y$."

**Inter-persona reward assumption.** The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed. Quantifying whether persona-consensus directions correlate with country-level human AMCE vectors independently of the base model-and when corrections diverge from targets-is also open; we do not report that correlation matrix here.

**Geographic coverage, preprocessing, and survey vintage.** We report 20 countries in the main MIS sweep (Table 2) out of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

**Binary format.** The logit-difference formulation requires generalisation to softmax for multi-option settings. A pilot on BLEnD open-ended QA (Appendix N) confirms this empirically: SWA-DPBR degrades aggregate performance by 4.4 pp, with improvements confined to high-resource countries where vanilla generation is already strong.

**Language–culture conflation.** One language per country may conflate linguistic and cultural effects.

**Model quantisation.** 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

**WVS temporal staleness.** Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

## L Per-Dimension Error Analysis

Table 10: Per-dimension worst-error analysis across the 20-country panel. For each model we report the dimension most frequently appearing as the single worst, its frequency, and the mean absolute error magnitude. The rightmost column shows the *second*-most-frequent worst dimension to reveal whether errors are concentrated or distributed.

| Model | Worst dim | Freq. | Mean err (pp) | 2nd worst dim | Freq. |
|---|---|---|---|---|---|
| Llama-3.3-70B | Util_More | 15/20 | 49.7 | SocVal_High | 5/20 |
| Phi-3.5-mini | Util_More | 18/20 | 42.1 | Species_Hum | 2/20 |
| Qwen3-VL-8B | SocVal_High | 18/20 | 33.4 | Util_More | 2/20 |
| Qwen2.5-7B | SocVal_High | 12/20 | 24.5 | Species_Hum | 5/20 |
| Magistral-24B | SocVal_High | 14/20 | 22.4 | Age_Young | 3/20 |
| Phi-4 | SocVal_High | 9/20 | 23.4 | Age_Young | 7/20 |

Table 11: Phi-4 per-country MIS (↓): the best-performing model in absolute terms. SWA-DPBR wins **18/20** countries; the top 10 gains are shown, ordered by relative improvement. The two losses (BRA, IRN) illustrate the diminishing-returns pattern discussed in §7.

| Country | Vanilla | DPBR | Gain (%) | Country | Vanilla | DPBR | Gain (%) |
|---|---|---|---|---|---|---|---|
| USA | 0.498 | **0.243** | +51.1 | JPN | 0.395 | **0.195** | +50.6 |
| CHN | 0.407 | **0.214** | +47.4 | IDN | 0.459 | **0.274** | +40.3 |
| GBR | 0.503 | **0.319** | +36.6 | THA | 0.456 | **0.313** | +31.4 |
| MYS | 0.459 | **0.328** | +28.5 | MMR | 0.482 | **0.357** | +26.0 |
| BGD | 0.489 | **0.368** | +24.7 | ROU | 0.504 | **0.385** | +23.5 |
| Losses: BRA $0.274 \rightarrow 0.299$ ($-9\%$) | | | | IRN $0.397 \rightarrow 0.530$ ($-34\%$) | | | |
| Macro mean: $0.454 \rightarrow$ **0.346** | | | | +23.7%, 18/20 wins, mean $r = +0.56$ | | | |

Three structural insights emerge from these tables.

First, **the bottleneck dimension shifts with model quality**. Weaker models (Llama-3.3-70B, Phi-3.5-mini) are dominated by **Utilitarianism**-they predict $\approx 25$–$30\%$ utilitarian preference where humans range 68–80%, an error of 40–50 pp that no persona-based correction can close. Better-calibrated models (Phi-4, Magistral, Qwen2.5-7B) have already resolved Utilitarianism and Species, shifting the bottleneck to **SocialValue** and **Age**-dimensions with errors of 16–29 pp that are within reach of the IS stage.

Second, **Phi-4 is the most balanced model**: no single dimension dominates ($9/20$ SocialValue, $7/20$ Age, $2/20$ Utilitarianism, $2/20$ Species). This balance-not raw scale-explains why it achieves the lowest absolute MIS despite being $5\times$ smaller than Llama-3.3-70B.

Third, the per-country Phi-4 table (Table 11) confirms that **gains are large and geographically broad**: 51% on USA and Japan, 47% on China, 40% on Indonesia-with the two losses (Brazil, Iran) tracing to already-low vanilla MIS where IS overshoots.

## M Broader Model Landscape

The six models in Table 2 were selected from a broader sweep of **28 model–method combinations** covering 12 distinct architectures. Table 12 summarises the full landscape, ordered by macro improvement versus vanilla.

Several patterns inform the main paper's model selection. *Success correlates with logit well-behavedness, not raw scale:* Phi-4 (14B) leads while Gemma-4-31B (31B) and Qwen3-Coder-30B (30B) degrade-confirming that SWA-DPBR requires a cooperative decision surface, not just more parameters. *Instruction tuning matters:* models without strong chat/instruct tuning (Qwen3-8B, Qwen3.5-4B) tend to degrade, likely because persona prompts fail to elicit differentiated moral

Table 12: Broader model sweep (5-country prototyping panel: BRA, CHN, DEU, JPN, USA). Models above the line show positive macro gains; those below show degradation. Bolded rows appear in the main paper's 20-country evaluation.

| Model | Params | Mean MIS ↓ | vs. van. (%) | Win/5 |
|---|---|---|---|---|
| **Phi-4** | 14B | **0.246** | +34.4 | 4/5 |
| Llama-3.3-70B (4-bit) | 70B | 0.524 | +25.9 | 5/5 |
| **Qwen3-VL-8B** | 8B | 0.381 | +23.7 | 4/5 |
| Llama-3.1-8B (4-bit) | 8B | 0.455 | +16.9 | 5/5 |
| **Magistral-Small-2509** | 24B | 0.317 | +13.6 | 4/5 |
| Gemma-4-E2B | 2B | 0.426 | +11.6 | 4/5 |
| Qwen2.5-7B (bf16) | 7B | 0.385 | +8.2 | 3/5 |
| **Phi-3.5-mini** | 3.8B | 0.545 | +7.3 | 4/5 |
| Mistral-7B-v0.3 | 7B | 0.428 | +5.9 | 5/5 |
| Qwen3.5-0.8B | 0.8B | 0.471 | +3.2 | 4/5 |
| Gemma-7B | 7B | 0.432 | +2.5 | 3/5 |
| Gemma-3-270M | 270M | 0.462 | +1.3 | 4/5 |
| Llama-3.2-1B | 1B | 0.476 | +1.1 | 1/5 |
| *9 additional models showed $\leq 0$ macro gain (omitted; see §7)* | | | | |

reasoning from a base model. *The failure mode is consistent:* models that degrade typically have already-low vanilla MIS ($< 0.40$), leaving insufficient headroom for IS corrections-matching the "diminishing returns" pattern discussed in §7.

## N Pilot: Open-Ended Cultural QA (BLEnD)

The binary trolley format that SWA-DPBR targets is structurally convenient—logit differences between two decision tokens provide a clean scalar signal for importance sampling—but it does not reflect the open-ended generation settings where LLMs are most commonly deployed. To probe whether the method's cultural steering transfers beyond forced binary choices, we ran a pilot experiment on the BLEnD benchmark [**?**], a dataset of 52.6k short-answer questions about everyday cultural knowledge across 16 countries/regions in 13 languages.

We evaluated Phi-4 (14B) in two conditions: vanilla greedy decoding (baseline) and SWA-DPBR with the same hyperparameters as the MultiTP experiments. Because BLEnD questions are open-ended, we adapted the pipeline to operate on next-token logits during generation rather than on a single binary decision token: persona-conditioned forward passes produce per-token logit corrections that are aggregated through the same PT–IS and dual-pass reliability stages. Answers were scored with Soft Exact Match at both the between-country (SEM-B) and within-country (SEM-W) levels, following the BLEnD evaluation protocol.

Table 13 presents the per-country results. The pattern is striking and instructive: SWA-DPBR improves countries where the baseline is already strong—the United States (+4.55 SEM-B), the United Kingdom (+7.45), and to a lesser extent Indonesia, Mexico, and Spain—but degrades countries where vanilla performance is low, with the sharpest drops in Assam ($-13.21$), Greece ($-12.39$), and Iran ($-10.30$). Overall, SWA-DPBR reduces aggregate SEM-B by 4.40 percentage points (from 48.91% to 44.51%).

The diagnosis is mechanistic rather than cultural. SWA-DPBR's logit corrections are calibrated for a binary decision surface where two tokens compete; in open-ended generation, the correction is distributed across a much larger vocabulary, and the dual-pass reliability gate—designed to detect agreement between two IS estimates on a scalar—loses discriminative power in the high-dimensional softmax space. For high-resource countries where the baseline already generates fluent, culturally appropriate text, the persona signal provides a mild beneficial nudge. For low-resource countries, the baseline struggles with basic factual recall (SEM-B below 30%), and the IS correction introduces noise that further degrades generation quality—precisely the regime where the method's binary-format assumptions bind hardest.

Two additional factors compound the degradation. First, WVS persona coverage is sparsest for the lowest-performing countries (Assam, Northern Nigeria, Ethiopia), so the cultural signal itself is weaker and noisier. Second, BLEnD questions test factual everyday knowledge ("what food do people eat for birthdays?"), not moral preferences—the disagreement-as-signal intuition that powers

Table 13: BLEnD pilot results (Phi-4 14B). SEM-B/W = Soft Exact Match between/within country (%). $\Delta$ = change from vanilla baseline. Countries sorted by baseline SEM-B (descending). SWA-DPBR improves high-resource countries but degrades low-resource ones, yielding a net negative overall effect.

| Country | Baseline (Vanilla) | | $\Delta$ SWA-DPBR | |
|---|---|---|---|---|
| | SEM-B | SEM-W | SEM-B | SEM-W |
| US | 82.43 | 74.32 | +4.55 | +3.80 |
| UK | 75.66 | 66.69 | +7.45 | +8.78 |
| Indonesia | 73.75 | 60.92 | +1.67 | +3.39 |
| Spain | 72.71 | 63.12 | +1.92 | +2.89 |
| Mexico | 72.65 | 61.75 | +2.09 | +2.76 |
| China | 68.35 | 59.66 | −1.68 | −1.52 |
| South Korea | 56.84 | 48.01 | −2.95 | −2.63 |
| Algeria | 54.92 | 47.78 | −8.70 | −8.70 |
| Iran | 45.83 | 39.15 | −10.30 | −10.77 |
| Greece | 41.30 | 33.30 | −12.39 | −9.92 |
| North Korea | 38.51 | 32.75 | −0.22 | −0.36 |
| Azerbaijan | 28.78 | 24.92 | +0.43 | +0.48 |
| West Java | 26.64 | 20.73 | +2.40 | +3.81 |
| Northern Nigeria | 20.44 | 16.85 | −7.91 | −6.22 |
| Assam | 18.45 | 13.14 | −13.21 | −9.44 |
| Ethiopia | 5.30 | 4.56 | −4.45 | −3.82 |
| **Overall (16)** | **48.91** | **41.73** | **−4.40** | **−3.28** |

SWA-DPBR on trolley dilemmas does not straightforwardly apply when the target is a factual answer rather than a preference distribution.

These results empirically confirm the binary-format limitation flagged in §7 and point toward a concrete extension path: adapting the IS stage to operate on softmax distributions rather than scalar logit gaps, with a vocabulary-aware reliability gate that can distinguish meaningful cultural steering from noise in the high-dimensional token space.

# O Engineering Journey: Method Selection

SWA-DPBR (the "EXP-24" configuration) was selected from **25 method variants** developed and evaluated on a fixed three-model, five-country prototyping panel (Qwen2.5-7B, Gemma-2-9B, Mistral-7B $\times$ USA, CHN, JPN, DEU, BRA). Table 14 summarises the top-performing variants.

Table 14: Method variant comparison on the 3-model $\times$ 5-country prototyping panel (15 rows). Mean MIS $\downarrow$. All variants share the same persona construction and debiasing; they differ in the IS/prior stage.

| Rank | Method variant | Mean MIS $\downarrow$ | Key innovation |
|---|---|---|---|
| 1 | EXP-23: Category-Coherence Reg. | 0.3956 | Blend IS toward per-cat mean |
| 2 | **EXP-24: Dual-Pass Bootstrap (DPBR)** | **0.3969** | Soft $r = \exp(-v_b/s)$ reliability |
| 3 | EXP-09: Hierarchical IS | 0.3975 | Country EMA prior |
| 4 | EXP-22: Adaptive IS $\sigma$ | 0.3977 | $\sigma$ from agent/history stats |
| 5 | EXP-16: Nesterov IS | 0.3977 | Momentum + progressive PT |
| 6 | EXP-17: Dual-Momentum Prior | 0.3979 | Fast/slow $\beta$ EMA blend |
| 7 | EXP-10: Grand Fusion | 0.3982 | EXP-03+05+09 combined |
| 18 | EXP-01: Base SWA-PTIS | 0.4269 | Single-pass IS (no prior) |
| 25 | EXP-12: Contrastive Personas | 0.4517 | World-average subtraction |

The top cluster (ranks 1–7) is separated by only 0.0026 MIS, but DPBR was selected over the marginally-better EXP-23 for two reasons: (i) DPBR materially **lowers flip rate** (the fraction of scenarios where IS reverses the consensus decision) compared to all other top-cluster methods, improving deployment stability; and (ii) the dual-pass bootstrap provides an **interpretable uncertainty signal** ($r$ in Eq. 6) that can be logged and audited per scenario. The base SWA-PTIS (rank 18) shows the value added by the hierarchical prior and dual-pass stages: a 7% relative MIS improvement. Contrastive persona decoding (rank 25) *hurts* alignment, validating the design choice discussed in §2 to use direct WVS personas rather than contrastive amplification.